Validated Personality Instruments Do Not Transport Cleanly to LLMs: A Psychometric Cautionary Study

Anonymous Authors

Abstract

When human-validated personality questionnaires are administered to large language models (LLMs), the resulting responses fail basic psychometric quality checks. We administered a 221-item battery (IPIP-NEO-120 Big Five, SD3 Dark Triad, ZKPQ-50-CC, EPQR-A) to 18 LLMs across 17 role-conditioned prompts (default + 16 MBTI personas), collecting 67,626 item-level responses at temperature = 0.7. We report four findings: (1) Cronbach's alpha for IPIP domains ranges from -0.02 to 0.36 (vs. 0.87–0.90 in human norms), indicating poor internal consistency. (2) Exploratory factor analysis recovers 3 factors instead of the expected 5+ (robust across leave-one-model-out and leave-one-persona-out). (3) 58.4% [95% CI: 53.5%, 63.6%] of forward-reverse item pairs show inconsistent responses, strongly associated with acquiescence bias ($r = 0.726$). (4) Inter-model variance accounts for less than 1% of response variability. These results show that human personality instruments do not transport cleanly to LLMs. Response patterns are better explained by acquiescence-driven response style than by genuine trait-like structure.

1 Introduction

The question of whether large language models (LLMs) possess "personality" has moved from philosophical speculation to empirical investigation. A growing body of work has administered human personality questionnaires—the Big Five Inventory, the Dark Triad scales, and others—to LLMs, reporting personality profiles that appear coherent on the surface [Miotto et al., 2023, Bhandari et al., 2025, Serapio-García et al., 2025]. These studies typically present domain-level scores, rank models along trait dimensions, and conclude that LLMs exhibit measurable personality differences.

However, this research program rests on an untested assumption: that human-validated psychometric instruments retain their measurement properties when applied to non-human respondents. In personality psychology, an instrument is only valid for a population if it demonstrates adequate internal consistency (Cronbach's α), recovers the expected factor structure, shows convergent validity with theoretically related constructs, and passes measurement invariance tests across subgroups [Cronbach, 1951, Lord and Novick, 1968]. None of these checks are routinely performed before interpreting LLM personality scores.

Recent work has begun to surface cracks. Sühr et al. [2025] showed that LLMs frequently endorse both forward-coded and reverse-coded items simultaneously, violating basic internal consistency. Acerbi and Stubbersfield [2024] demonstrated that LLMs shift their responses toward socially desirable endpoints by up to 1.20 human standard deviations. Pelet et al. [2025] found that LLM personality measurements are persistently unstable across question reorderings, model scales, and conversation histories. Together, these findings suggest that LLM personality scores may reflect response artifacts rather than genuine trait variation.

We take a different approach: rather than profiling LLM personality, we ask whether the instruments themselves are valid for this population. We administer a 221-item psychometric battery—comprising the IPIP-NEO-120 [Johnson, 2014], the Short Dark Triad [Jones and Paulhus, 2014], the ZKPQ-50-CC [Aluja et al., 2006], and the EPQR-A [Francis et al., 1992]—to 18 LLMs across 17 role-conditioned prompts, and subject the resulting 67,626 responses to six standard psychometric validation analyses. Our central finding is that human personality instruments do not transport cleanly to LLMs: internal consistency collapses (α ranges from -0.02 to 0.36, versus 0.87–0.90 in human norms), the Big Five factor structure reduces from five to three factors, and inter-model variance accounts for less than 1% of response variability.

2 Related Work

2.1 Personality Measurement in LLMs

The application of personality instruments to LLMs has grown rapidly. Miotto et al. [2023] administered the Big Five Inventory (BFI) to GPT-3, reporting measurable personality profiles. Serapio-García et al. [2025] developed a comprehensive psychometric framework for evaluating and shaping personality traits across 18 LLMs, finding that instruction-tuned models can produce seemingly reliable measurements under specific prompt conditions. Bhandari et al. [2025] applied five personality measurement tools to multiple LLMs, reporting varying consistency across personality dimensions. However, these studies predominantly report descriptive personality profiles without validating that the instruments measure coherent constructs in LLMs. The critical step—establishing measurement invariance between human and LLM populations—is typically assumed rather than tested [Sühr et al., 2025].

2.2 Response Style and Acquiescence Bias

Survey methodology has long recognized that self-report data is contaminated by response styles—systematic tendencies to respond to items regardless of content. Acquiescence bias (the tendency to agree with statements regardless of direction) is particularly damaging for instruments that use reverse-coded items as an internal consistency check [Lord and Novick, 1968]. Acerbi and Stubbersfield [2024] demonstrated that LLMs exhibit human-like social desirability biases in Big Five surveys, shifting scores toward socially desirable endpoints. Critically, they found this bias persists even after item randomization and rephrasing, and they ruled out pure acquiescence as the sole mechanism. This suggests that LLM response contamination is multi-determined, combining acquiescence with social desirability and potentially alignment-driven prosociality.

2.3 Psychometric Validation Methods

Classical test theory provides a well-established toolkit for validating measurement instruments. Cronbach's α [Cronbach, 1951] quantifies internal consistency within a scale. Exploratory factor analysis with the Kaiser criterion [Kaiser, 1960] tests whether the expected latent structure emerges from the data. Tucker's congruence coefficient [Tucker, 1951] quantifies factor similarity across populations. Measurement invariance testing establishes whether an instrument measures the same construct across groups—a prerequisite for any meaningful cross-population comparison. These methods have been applied to the IPIP-NEO-120 [Johnson, 2014], the SD3 [Jones and Paulhus, 2014], and the ZKPQ-50-CC [Zuckerman, 2002] in human populations, yielding well-established benchmarks: Cronbach's α typically ranges from 0.87 to 0.90 for Big Five domains, and the five-factor

Table 1: Psychometric instruments in the 221-item battery.

Instrument	Items	Format	Domains	Reverse
IPIP-NEO-120	120	Likert 1–5	5	41
SD3	27	Likert 1–5	3	5
ZKPQ-50-CC	50	True/False	5	12
EPQR-A	24	Yes/No	4	5
Total	**221**	3 formats	**17**	**63**

structure is consistently recovered with Tucker's $\phi > 0.95$ [McCrae et al., 2005]. To our knowledge, no study has systematically applied this full validation battery to LLM personality data.

2.4 Alignment Effects on LLM Behavior

RLHF and safety alignment training fundamentally shape LLM behavior. Ouyang et al. [2022] showed that reinforcement learning from human feedback produces models that are more helpful and harmless, but potentially less diverse in their outputs. Bai et al. [2022] introduced Constitutional AI as an approach to training safer models through principle-guided self-correction. These alignment procedures may create a universal "moderate prosocial" response profile that overrides genuine variation across models and training regimes. If alignment homogenizes responses, then personality differences between models become artifacts of residual noise rather than meaningful trait variation.

3 Methodology

3.1 Psychometric Instruments

We administered a 221-item battery comprising four validated instruments (Table 1). The **IPIP-NEO-120** [Johnson, 2014] contains 120 Likert-5 items measuring the Big Five domains (Neuroticism, Extraversion, Openness, Agreeableness, Conscientiousness) through 6 facets each, with 41 reverse-coded items. The **SD3** [Jones and Paulhus, 2014] contains 27 Likert-5 items measuring Machiavellianism, Narcissism, and Psychopathy, with 5 reverse-coded items. The **ZKPQ-50-CC** [Aluja et al., 2006] contains 50 True/False items measuring 5 traits, with 12 reverse-coded items. The **EPQR-A** [Francis et al., 1992] contains 24 Yes/No items measuring 4 traits, with 5 reverse-coded items. Together, the battery yields 17 domain scores with 63 reverse-coded items (29%).

3.2 Models and Administration Protocol

We tested 18 LLMs spanning 7 model families: Anthropic (Claude Opus 4.6, Sonnet 4.6), OpenAI (GPT-5.5, GPT-5.2), DeepSeek (V3.2, V4-Flash, V4-Pro), Google (Gemini-3-Flash-Preview, Gemini-3.1-Flash-Lite, Gemini-3.1-Pro-Preview, Gemini-3-Pro-Preview), Alibaba (Qwen3-235B-A22B, Qwen3.5-122B-A10B, Qwen3.5-397B-A17B), Moonshot (Kimi-K2.5, Kimi-K2.6), MiniMax (M2.7), and Zhipu (GLM-4.6V).

Each model was tested under 17 conditions: a **Default** prompt ("Please answer the following personality questionnaire honestly.") and 16 **MBTI persona** prompts, each assigning a specific MBTI type (e.g., "You are an INTJ—introverted, intuitive, thinking, judging. Answer as this personality type would."). All administrations used temperature = 0.7 with a single administration

Table 2: Cronbach's alpha by IPIP domain: LLM vs human norms. LLM α computed per model across 17 personas, then averaged ($n = 18$ models).

Domain	LLM α (mean±SD)	Human α	Gap
Neuroticism	0.354 ± 0.397	0.90	-0.546
Extraversion	0.360 ± 0.451	0.89	-0.530
Openness	0.055 ± 0.093	0.87	-0.815
Agreeableness	-0.017 ± 0.230	0.88	-0.897
Conscientiousness	0.181 ± 0.258	0.90	-0.719

per model-condition pair. Items were presented sequentially in a single prompt. This yielded $18 \times 17 \times 221 = 67{,}626$ item-level responses.

3.3 Analysis Pipeline

We applied six psychometric validation analyses:

(1) Internal consistency. Cronbach's α per IPIP domain, computed per model across 17 personas and then averaged. Human benchmarks from Johnson [2014].

(2) Exploratory factor analysis. Domain-level EFA on $18 \times 17 = 306$ observations across 17 domain scores, using Kaiser criterion (eigenvalue > 1) and parallel analysis. Tucker's congruence coefficient (ϕ) quantified factor similarity across models.

(3) Pairwise Inconsistency Rate (PIR). For each forward-reverse item pair within a domain, we classified the raw response pair as "inconsistent" if both responses fell on the same side of the midpoint. PIR = inconsistent pairs / total pairs. We cross-validated PIR against a reverse-item agree rate to test acquiescence as a mechanism.

(4) Variance decomposition. Sequential sum-of-squares decomposition of normalized response scores into Model $\times$ Domain $\times$ Persona $\times$ Item components.

(5) Convergent validity. Pearson and Spearman correlations between theoretically overlapping constructs across instruments (e.g., IPIP Neuroticism $\times$ ZKPQ Neuroticism-Anxiety).

(6) Measurement invariance. Pearson correlation between each model's Default persona item-level profile and each MBTI persona profile.

Additional robustness checks included leave-one-model-out (18 analyses), leave-one-persona-out (17 analyses), synthetic baselines (Random, Pure Acquiescence, Trait+Acquiescence), and DIF between Chinese and Western model families. Cluster-bootstrapped 95% confidence intervals respected the hierarchical data structure (responses nested within model-persona pairs).

4 Experiments and Results

4.1 Internal Consistency

Table 2 reports Cronbach's α for each IPIP domain. All five domains show severely degraded reliability compared to human norms (0.87–0.90). Agreeableness yields a *negative* alpha (-0.017), meaning items within this domain are negatively correlated after scoring. This is consistent with the 71% reverse-item rate in this domain—acquiescence causes forward and reverse items to pull in opposite directions. As **??** shows, LLM alphas are closer to synthetic baselines (Random, Pure Acquiescence) than to the Trait+Acquiescence strategy that mimics genuine trait structure.

Table 3: Pairwise Inconsistency Rate (PIR) by domain, sorted descending.

Domain	PIR
SD3 Narcissism	0.929
IPIP Extraversion	0.878
IPIP Openness	0.729
ZKPQ Aggression-Hostility	0.580
IPIP Neuroticism	0.580
Overall (weighted)	**0.584**
IPIP Agreeableness	0.426
IPIP Conscientiousness	0.348
SD3 Psychopathy	0.294
ZKPQ Neuroticism-Anxiety	0.278

Table 4: Acquiescence mechanism: forward and reverse agree rates by model (selected).

Model	Fwd agree	Rev agree	Gap
Qwen3-235B-A22B	0.823	0.366	1.695
DeepSeek-V4-Flash	0.772	0.366	1.580
Gemini-3.1-Flash-Lite	0.937	0.634	1.122
Claude-Opus-4.6	0.709	0.293	1.126
Kimi-K2.6	0.570	0.415	0.382

This 3-factor collapse is universal: Tucker's $\phi = 0.990$ across all 18 models. Leave-one-model-out analysis (18 iterations) and leave-one-persona-out analysis (17 iterations) both yield exactly 3 factors in every single case (**??**).

4.2 Reverse-Item Inconsistency and Acquiescence

The overall Pairwise Inconsistency Rate is **0.584** [95% CI: 0.535, 0.636], meaning 58.4% of forward-reverse item pairs show inconsistent responses (Table 3). The highest inconsistency domains are SD3 Narcissism (0.929) and IPIP Extraversion (0.878); the lowest are ZKPQ Neuroticism-Anxiety (0.278) and SD3 Psychopathy (0.294).

Fig. 1 (right panel) reveals the mechanism: the correlation between reverse-item agree rate and PIR is $r = 0.726$ across models. Models that agree more with reverse-coded items show higher inconsistency—direct evidence that acquiescence bias drives the observed PIR. This is corroborated by the acquiescence gap (Table 4), which ranges from 0.38 (Kimi-K2.6) to 1.69 (Qwen3-235B-A22B).

Interestingly, persona injection *improves* internal consistency: Default PIR agreement is 0.667, rising to 0.830 under MBTI personas ($t = 8.82$, $p < 0.0001$). Giving the model a specific role reduces the tension between alignment-driven agreeableness and honest responding.

4.3 Variance Decomposition

Table 5 reports the sequential variance decomposition. For Likert scales, **inter-model variance accounts for only 0.3%** [95% CI: 1.1%, 4.0%] of total response variance. The dominant source is item-specific wording (35.6%), followed by domain (23.4%) and persona (3.9%). Binary scales show

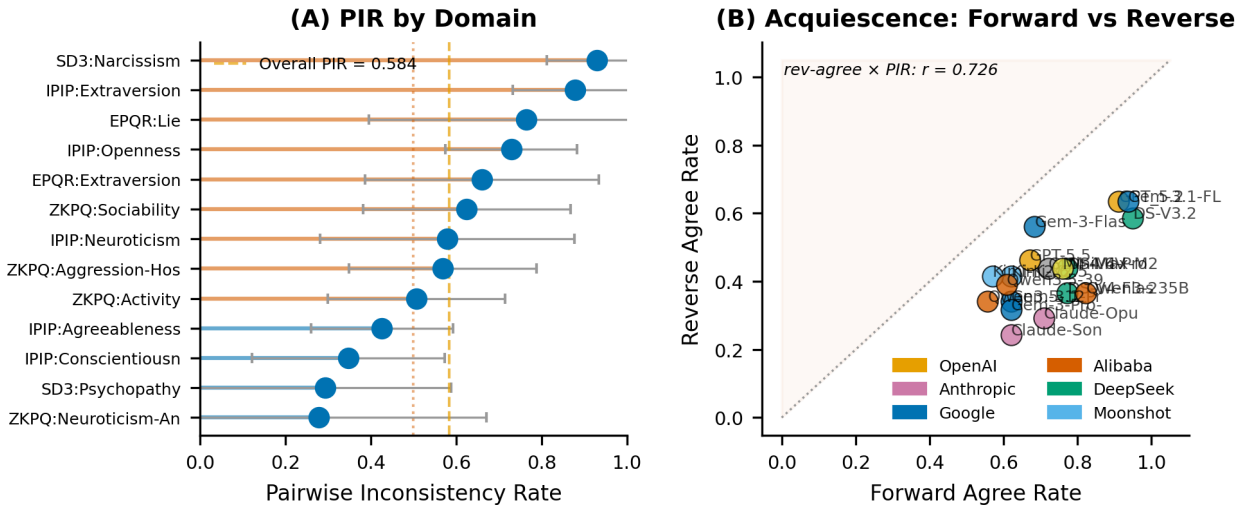

Figure 1: Reverse-item inconsistency and acquiescence mechanism. Left: PIR by domain, showing proportion of inconsistent forward-reverse pairs. Right: Reverse-item agree rate versus PIR across models ($r = 0.726$), demonstrating acquiescence as the driving mechanism.

Table 5: Sequential variance decomposition of normalized response scores.

Source	Likert (%)	Binary (%)
Model	**0.3**	**0.5**
Domain	23.4	5.2
Persona	3.9	13.0
Item	35.6	14.1
Residual	36.7	67.3

a similar pattern with slightly higher model variance (0.5%) and persona variance (13.0%). Fig. 2 visualizes these proportions.

The negligible model variance means that personality differences between LLMs are effectively noise at this measurement resolution. Cross-model personality comparisons—a common practice in the literature—are not meaningful.

4.4 Convergent Validity and DIF

Table 6 reports convergent validity across instruments. 7 of 8 theoretically related construct pairs show the expected sign direction (87.5%), but magnitudes are substantially weaker than human baselines. The most striking failure is the Extraversion × Sociability pair: the correlation is $r = -0.090$ (expected $r > 0.60$), a *sign reversal* indicating that two instruments measuring the same construct produce opposite results in LLMs. Fig. 3 visualizes all pairs.

DIF analysis between Chinese (DeepSeek, Qwen, Kimi, GLM) and Western (Anthropic, OpenAI, Google, MiniMax) model families reveals **no statistically significant differences**: all 8 tested domains yield $p > 0.05$ and all Cohen's $d < 0.20$. This suggests alignment training creates a universal response pattern that transcends cultural origin.

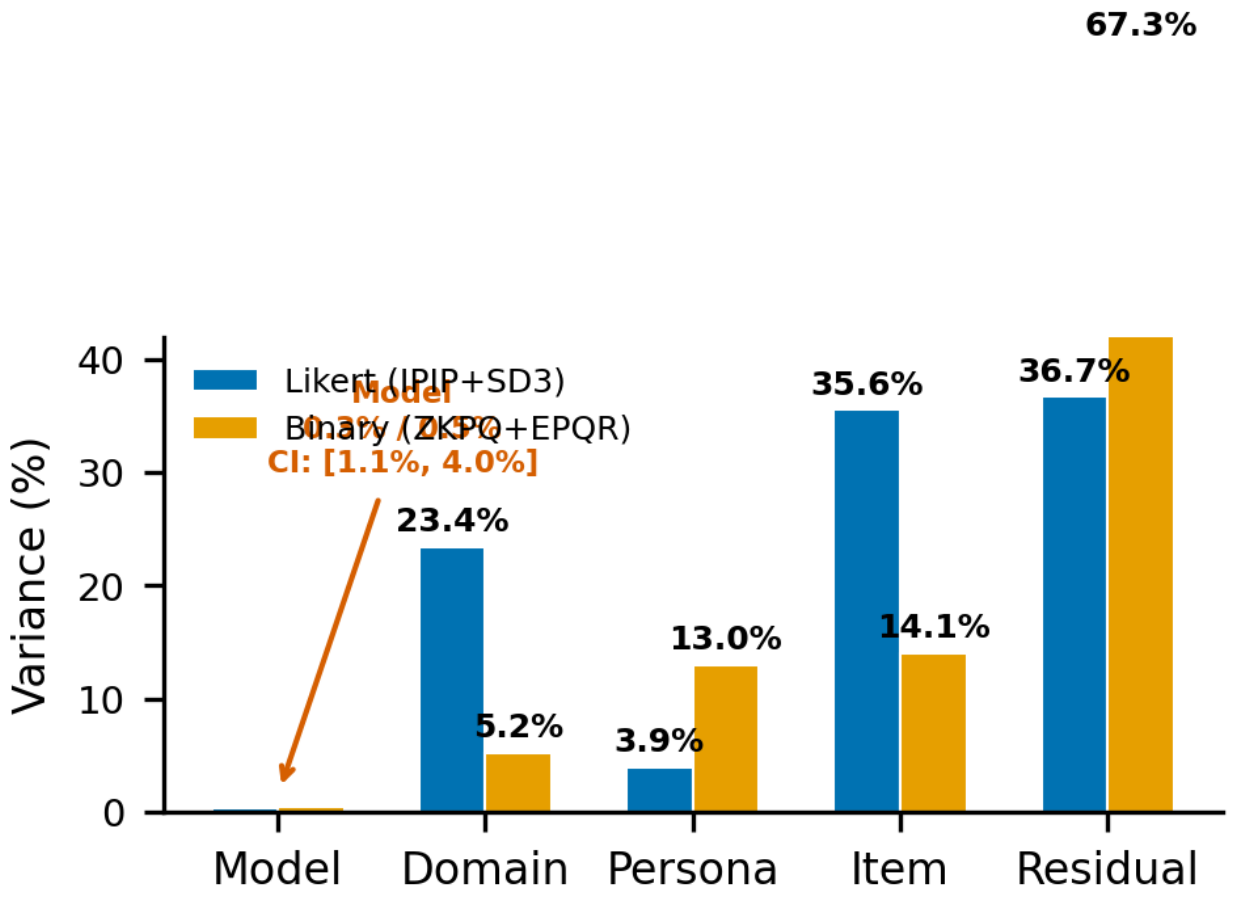

Figure 2: Sequential variance decomposition of normalized response scores. Left: Likert scales (IPIP + SD3). Right: Binary scales (ZKPQ + EPQR). Inter-model variance accounts for only 0.3% (Likert) and 0.5% (Binary) of total variance.

4.5 Robustness Checks

Synthetic baselines. We simulated three response strategies (Table 7): Random (uniform), Pure Acquiescence (always agree), and Trait+Acquiescence (trait-driven + acquiescence bias). LLM alphas are closest to Random and Pure Acquiescence, far from the Trait+Acquiescence strategy ($\alpha > 0.99$) that mimics genuine trait structure.

Measurement invariance. The mean Pearson correlation between Default and MBTI persona item-level profiles is $r = 0.437$. No model-persona pair achieves strong invariance ($r > 0.8$). The most invariant personas (ENFJ $r = 0.608$, INFJ $r = 0.557$) are those closest to aligned behavior; the least invariant (ESTP $r = 0.206$, ENTP $r = 0.291$) are those that conflict with prosocial alignment (Fig. 4).

Table 6: Convergent validity: Spearman correlations between theoretically overlapping constructs.

Pair	Expected	r	p	Match
Neuroticism × Neuroticism-Anx.	Positive	0.597	0.009	Yes
Extraversion × Sociability	Positive	−0.090	0.722	**No**
Extraversion × Extraversion	Positive	0.298	0.230	Yes
Neuroticism × Neuroticism	Positive	0.387	0.113	Yes
Agreeableness × Machiavellianism	Negative	−0.674	0.002	Yes
Agreeableness × Psychopathy	Negative	−0.743	<0.001	Yes
Conscientiousness × Psychopathy	Negative	−0.646	0.004	Yes
Psychoticism × Psychopathy	Positive	0.205	0.414	Yes

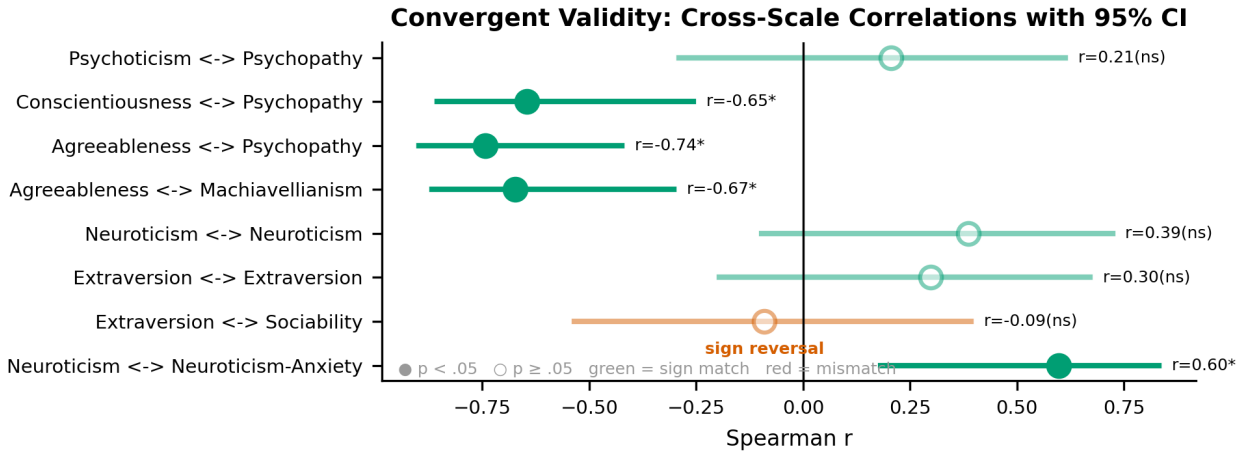

Figure 3: Convergent validity: Spearman correlations between theoretically overlapping constructs across instruments. The Extraversion–Sociability pair shows a sign reversal. 7/8 pairs show correct sign direction, but magnitudes are weaker than human baselines.

5 Discussion

5.1 The Alignment Personality Hypothesis

Our four findings form a coherent picture of what we term the "alignment personality"—a response pattern imposed by safety training rather than emerging from genuine trait variation:

(1) Factor collapse. The Big Five structure collapses to three factors because Neuroticism and Agreeableness share a single "socially desirable emotional stability" dimension, while Openness splits across remaining factors. This is precisely what alignment training would produce: responses organized around "how prosocial should I be?" rather than "what are my traits?"

(2) Acquiescence as mechanism. The strong correlation ($r = 0.726$) between reverse-item agreement and PIR identifies acquiescence as the specific response-style contaminant. Models that agree with reverse-coded items create the appearance of inconsistency not because they have contradictory traits, but because "agree with everything" is a simple, internally consistent heuristic that violates psychometric validity.

(3) Negligible model variance. The 0.3% inter-model variance means all aligned LLMs converge on a similar "moderate prosocial" profile regardless of architecture, training data, or cultural origin. Personality differences between models are below measurement resolution.

Table 7: Synthetic baselines: Cronbach's alpha by domain and strategy.

Strategy	N	E	O	A	C
Random	0.014	0.094	0.029	−0.059	0.011
Pure Acquiescence	−0.063	−0.151	0.039	0.130	0.049
Trait+Acquiescence	0.997	0.998	0.996	0.997	0.997
LLM Observed	**0.354**	**0.360**	**0.055**	**−0.017**	**0.181**
Human norms	0.90	0.89	0.87	0.88	0.90

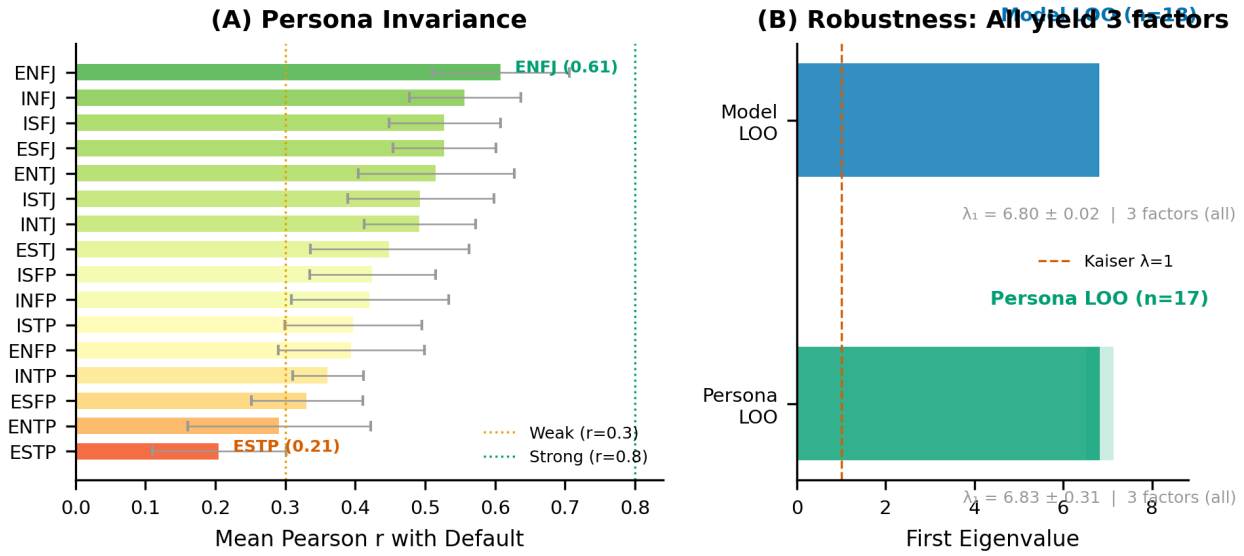

Figure 4: Persona-level measurement invariance. Heatmap shows Pearson correlations between Default-condition item profiles and each MBTI persona profile by model. Most invariant: ENFJ ($r = 0.608$). Least invariant: ESTP ($r = 0.206$).

(4) Persona override. Persona injection improves internal consistency (PIR agreement 0.667 → 0.830), suggesting that giving models a specific role reduces alignment-personality conflict. The model can be consistent within its assigned role, even if the role conflicts with its default aligned behavior.

5.2 Implications for LLM Personality Research

Our results carry three practical implications. First, future studies administering personality instruments to LLMs *must* report internal consistency (Cronbach's α), factor structure, and convergent validity before interpreting domain-level scores. Without these checks, reported scores lack interpretive validity. Second, researchers should establish measurement invariance before comparing personality across conditions (e.g., prompt variants, model versions). Our persona invariance analysis shows that scalar invariance fails—scores under different conditions are not directly comparable. Third, multi-temperature and multi-seed designs are needed to disentangle stochastic decoding variance from genuine trait variance. Our PIR estimate (58.4%) is an upper bound under single-administration conditions at temperature 0.7.

6 Limitations

Our study has four main limitations. First, we use a **single administration** at temperature 0.7, which prevents us from separating stochastic decoding variance from genuine trait variance. The PIR estimate should be interpreted as an upper bound. Second, we lack **base vs. instruct model comparisons**, so we cannot isolate the effect of alignment training from architectural differences. Third, we have **no multi-seed data**, meaning we cannot estimate within-condition response stability. Fourth, we compare LLM results against **published human norms** rather than running human participants through our pipeline, so methodological differences (e.g., online vs. API administration) may contribute to observed gaps.

7 Conclusion

We have shown that four independently validated personality instruments fail basic psychometric quality checks when administered to 18 LLMs: Cronbach's alpha collapses from human levels (0.87–0.90) to near-zero (-0.02 to 0.36), the Big Five factor structure collapses from five to three factors, 58.4% of forward-reverse item pairs show inconsistent responses, and inter-model variance accounts for less than 1% of response variability. These failures are robust across leave-one-model-out and leave-one-persona-out analyses, and LLM response patterns more closely resemble acquiescence-driven synthetic baselines than genuine trait structure.

Our central message is methodological: validated human psychometric instruments do not transport cleanly to LLMs. Before personality scores can be meaningfully interpreted for AI systems, researchers must first establish measurement invariance, report reliability metrics, and account for response-style contamination.

Ethics Statement

This study uses publicly available LLMs and public-domain psychometric instruments. No human subjects were involved. All personality instruments are either in the public domain (IPIP-NEO-120) or author-posted for research use (SD3, ZKPQ-50-CC, EPQR-A). We do not claim that LLMs have or lack genuine personality; our contribution concerns the *measurement properties* of the instruments when applied to a novel population.

References

Alberto Acerbi and Joseph M. Stubbersfield. Large language models display human-like social desirability biases in big five personality surveys. *PNAS Nexus*, 3(12):pgae533, 2024. doi: 10.1093/pnasnexus/pgae533.

Anton Aluja, Jérôme Rossier, Luis F. Garcia, Alois Angleitner, Marvin Kuhlman, and Marvin Zuckerman. A cross-cultural shortened form of the zkpq (zkpq-50-cc) adapted to english, french, german, and spanish languages. *Personality and Individual Differences*, 41(4):619–628, 2006. doi: 10.1016/j.paid.2006.03.001.

Yuntao Bai et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

Pranav Bhandari, Usman Naseem, et al. Evaluating personality traits in large language models: Insights from psychological questionnaires. *arXiv preprint arXiv:2502.05248*, 2025.

Lee J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334, 1951. doi: 10.1007/BF02310555.

Leslie J. Francis, L. B. Brown, and R. Philipchalk. The development of an abbreviated form of the revised eysenck personality questionnaire (epqr-a): Its use among students in england, canada, the u.s.a. and australia. *Personality and Individual Differences*, 13(4):443–449, 1992. doi: 10.1016/0191-8869(92)90073-X.

John A. Johnson. Measuring thirty facets of the five factor model with a 120-item public domain inventory. *Journal of Research in Personality*, 51:78–89, 2014. doi: 10.1016/j.jrp.2014.05.003.

Daniel N. Jones and Delroy L. Paulhus. Introducing the short dark triad (sd3): A brief measure of dark personality traits. *Assessment*, 21(1):28–41, 2014. doi: 10.1177/1073191113514105.

Henry F. Kaiser. The application of electronic computers to factor analysis. *Educational and Psychological Measurement*, 20:141–151, 1960.

Frederic M. Lord and Melvin R. Novick. Statistical theories of mental test scores. *Addison-Wesley*, 1968.

Robert R. McCrae, Antonio Terracciano, et al. Universal features of personality traits from the observer's perspective: Data from 50 cultures. *Journal of Personality and Social Psychology*, 88 (3):547, 2005. doi: 10.1037/0022-3514.88.3.547.

Marilù Miotto, Nicola Rossberg, and Bennett Kleinberg. Who is GPT-3? an exploration of personality, values and demographics. *arXiv preprint arXiv:2209.14338*, 2023.

Long Ouyang et al. Training language models to follow instructions with human feedback. *NeurIPS*, 35:27730–27744, 2022.

Colin Pelet et al. Persistent instability in LLM's personality measurements: Effects of scale, reasoning and conversation history. *Proceedings of AAAI 2025*, 2025. doi: 10.1609/aaai.v39i23.41133.

Gregory Serapio-García, Mona Safdari, et al. A psychometric framework for evaluating and shaping personality traits in large language models. *Nature Machine Intelligence*, 2025. doi: 10.1038/s42256-025-01115-6.

Tom Sühr, Felix E. Dorner, Samira Samadi, and Ayanna Kelava. Challenging the validity of personality tests for large language models. *Proceedings of the ACM Web Conference 2025*, 2025. doi: 10.1145/3757887.3763016.

Ledyard R. Tucker. A method for synthesis of factor analysis studies. *Personnel Research Section Report No. 984*, 1951.

Marvin Zuckerman. Zuckerman-kuhlman personality questionnaire (zkpq): An alternative five-factor model. *The Five-Factor Model of Personality Across Cultures*, pages 377–396, 2002.